

Kunhee (Geoffrey) Kim

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SUMMARY

UC Berkeley Data Science graduate and data engineer who builds and scales production data pipelines, multi-terabyte data lakes and warehouses (PostgreSQL, Neo4j, Spark), and ML/NLP systems. Strong in robust ETL with strict data-integrity guarantees, SQL performance tuning, and Linux infrastructure. Bilingual in English and Korean.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Swift, Bash; PostgreSQL, Neo4j, Apache Spark, PostGIS, SQLite

Data Engineering: ETL/ELT pipeline design, data lake & warehouse architecture, schema design, idempotent ingestion, deduplication, performance tuning, web scraping

Python & ML/NLP: Pandas, Polars, FastAPI, SQLAlchemy, Pydantic; local LLM inference (Ollama), Graph Neural Networks, entity resolution, backtesting

Tools & Languages: Docker, Linux, cron/systemd, Git, Tableau; English (fluent), Korean (fluent)

EXPERIENCE

Global Key Advisors — Financial Data Analyst

Jan 2026 – Present · Bay Area, CA

- Re-architected SEC EDGAR scraping pipelines from a memory-constrained CSV approach to a threaded PostgreSQL integration, scaling the data lake to **2.05M+ text filings and 3.3M+ financial records across 28,000+ entities**.
- Administered a PostgreSQL research warehouse holding **33M+ records and 42 GB of text payloads**; architected modular schemas, tuned TOAST storage, and debugged the query planner via system catalogs and ANALYZE statistics.
- Automated daily ingestion of a 1.12 GB price panel to a Linux Spark cluster with CIK-aligned joins, vendor deduplication, and idempotent upserts, **eliminating IP-ban risk and saving hours of manual alignment weekly**.
- Built NLP extraction with local LLMs (Ollama) parsing filings into corporate-relationship edges in a **Neo4j knowledge graph** via an async pipeline processing **~480 filings/hour** unattended.
- Migrated the prediction strategy from FinBERT sentiment to Graph Neural Networks; out-of-sample backtesting **exposed lookahead bias** (a +2.39% return collapsed to negative under 30-bps costs), preventing a losing strategy deployment.

PROJECTS

Alpha Signals — Quantitative SEC Event Research Pipeline

Global Key Advisors

Python, Polars, PostgreSQL, Neo4j, Ollama, Graph Neural Networks

- Built an autonomous daily pipeline (signals → simulated trades → GNN predictions by 04:30 PDT) evaluating whether SEC-filing corporate events predict stock returns over a Neo4j knowledge graph.

PathWise — Intelligent iOS Running App

Spring 2026

Swift (SwiftUI, MapKit, ActivityKit), FastAPI, PostgreSQL + PostGIS, GraphHopper, Docker

- Full-stack architect of a routing app with a GraphHopper + PostGIS orchestrator and native turn-by-turn navigation; achieved **sub-2-second route generation** and 100% uptime via a database-agnostic fallback. Presented at Berkeley's startup pitch competition.

EDUCATION

University of California, Berkeley — B.A. Data Science

May 2026

GPA 3.81 · Completed in 3 years · Berkeley Changemaker & Entrepreneurship & Technology (SCET) Certificates · 2-yr military service, Korea (2022–24)

Other experience: Student Researcher (Prof. Berrick, FamiliesNotFees) — ML on child-welfare data, Tableau dashboards, site grew to 4,000+ views; Course Coordinator, ENGIN 270B (100+ students); Berkeley Data Discovery — ML meta-analysis of education metrics.